Yukun Guo

Ph.D. Candidate

(503) 380-0669
guoyukn@gmail.com
Portland, OR
LinkedIn

SUMMARY/OBJECTIVE

Ph.D. Candidate specializing in Optical Coherence Tomography (OCT) and Artificial Intelligence in Ophthalmic Imaging. Over eight (8+) years of experience developing deep learning models for retinal image segmentation, image classification, image reconstruction, biomarker quantification, and disease progression analysis. Passionate about AI model design, exploration, and advancing machine vision technologies. Inventor of 3 U.S. patents and author of 40+ peer-reviewed publications. Expert in transforming advanced imaging algorithms into clinically usable software tools.

EDUCATION

2022 - Present: *Oregon Health & Science University*, Doctor of Philosophy, Biomedical Engineering

2014 - 2017: *University of Jinan*, Master of Science, Computer Science

2009 - 2013: *University of Jinan*, Bachelor of Science, Computer Science

SKILLS

- Deep learning, AI, machine learning
- Python, C++, MATLAB, JS/TS
- Scientific writing, Scientific presentation
- PyTorch, TensorFlow, Keras
- Microsoft Azure, GitHub
- Computer Vision, Medical Image Processing, Image Analysis, Algorithm
- Image Segmentation, Image Classification, Image Heigh-resolution Reconstruction, Signal Restoration
- Optical Coherence Tomography, Retinal Imaging
- Software development, Web application development

WORK EXPERIENCE

GRADUATE RESEARCHER

Oregon Health & Science University | Portland, OR | 2022 – Present

- Develops advanced deep learning frameworks for retinal OCT and OCTA image analysis, focusing on segmentation, biomarker quantification, and disease progression modeling for AMD and diabetic retinopathy.
- Designed multi-decoder convolutional neural networks for simultaneous segmentation of drusen, retinal pigment epithelium, and vascular structures, leading to AI-aided segmentation of four drusen types in volumetric OCT (Biomed. Opt. Express, 2025).
- Co-invented U.S. patents on automated detection of avascular and signal reduction regions in retinal OCTA (US11974811, 2024).
- Published over 15 papers from 2022–2025 in high-impact journals, including Ophthalmology Retina, Eye, Investigative Ophthalmology & Visual Science, and Biomedical Optics Express.
- Collaborates closely with clinical and imaging teams to translate AI algorithms into quantitative tools for clinical trials and early disease detection.

RESEARCH ASSISTANT

Oregon Health & Science University | Portland, OR | 2019 – 2022

- Built multiple end-to-end convolutional neural networks for OCTA image segmentation, including Automated Segmentation of Retinal Fluid Volumes from Structural and Angiographic OCT Using Deep Learning (TVST, 2020).
- Developed MEDnet, an AI system for automated detection of avascular areas in OCT angiography (Biomed. Opt. Express, 2018 – 2019), later adopted in clinical research for diabetic retinopathy monitoring.

- Designed deep learning models for wide-field OCTA quantification, resulting in publications such as Quantification of Nonperfusion Area in Montaged Widefield OCTA Using Deep Learning (Ophthalmology Science, 2021).
- Contributed to open-source retinal image reconstruction and segmentation tools, including An Open-Source Deep Learning Network for Reconstruction of High-Resolution OCT Angiograms (TVST, 2021).
- Published 20+ peer-reviewed papers (2019–2022) across Biomedical Optics Express, Investigative Ophthalmology & Visual Science, and Ophthalmology Retina, significantly advancing AI-OCT research reproducibility

VISITING RESEARCHER

Oregon Health & Science University | Portland, OR | 2017 – 2019

- Initiated collaborative research on automated OCTA analysis, leading to early publications on Automated Segmentation of Retinal Layer Boundaries and Capillary Plexuses (Biomed. Opt. Express, 2018).
- Developed early prototypes of deep learning algorithms for retinal vasculature segmentation that later evolved into patented clinical tools.

PATENTS

Detecting Avascular and Signal Reduction Areas in Retinas Using Neural Networks

US 11974811 B2 · Issued May 7, 2024

This disclosure describes systems, devices, and techniques for training neural networks to identify avascular and signal reduction areas of optical coherence tomography angiography images and for using trained neural networks.

Identifying retinal layer boundaries Identifying retinal layer boundaries

US 11717155 B2 · Issued Aug 8, 2023

Methods for automatically identifying retinal boundaries from reflectance optical coherence tomography images.

Detecting avascular areas using neural networks

US 11263747 B2 · Issued Mar 1, 2022

This method describes a deep learning approach for avascular area analysis using optical coherence tomography angiography images.

AWARDS

2021 Outstanding Reviewers

Optical Society of America (OSA), 2021

Travel Award

International Ocular Circulation Society (IOCS). Kyoto, Japan, 2021

Retina Research Foundation/Joseph M. and Eula C. Lawrence Travel Grant

Association for Research in Vision and Ophthalmology (ARVO), 2019

PEER-REVIEWED JOURNAL PAPERS

1. AI-aided segmentation of four types of drusen in volumetric OCT
Yukun Guo, Tristan T. Hormel, An-Lun Wu, Min Gao, Thomas S. Hwang, Steven T. Bailey and Yali Jia
Biomed. Opt. Express, Nov, 2025. 16(11): 4380--4391.

2. Wide-field OCTA quantified peripheral nonperfusion areas predict the risk of subclinical neovascularization
An-Lun Wu, **Yukun Guo**, Tristan T. Hormel, Christina J. Flaxel, Merina Thomas, Steven T. Bailey, Dong-Wouk Park, Yali Jia and Thomas S. Hwang
Eye, 8, 2025. 39(12): 2467--2473.

3. High-resolution multimodal visible light optical coherence tomography and scanning laser ophthalmoscopy for in vivo neuronal and vascular retinal imaging in mice
Siyu Song, Guangru Ben Liang, Tristan T. Hormel, **Yukun Guo**, Min Gao, Benjamin Sivyer, J. Peter Campbell, Siyu Chen, Yifan Jian and Yali Jia
Biomed. Opt. Express, Jun, 2025. 16(6): 2365--2375.

4. Nonperfused Retinal Capillaries—A New Method Developed on OCT and OCTA
Min Gao, **Yukun Guo**, Tristan T. Hormel, Jie Wang, Elizabeth White, Dong-Wouk Park, Thomas S. Hwang, Steven T. Bailey and Yali Jia
Investigative Ophthalmology & Visual Science, 04, 2025. 66(4): 22-22.

5. Widefield OCT angiography
Yali Jia, Tristan T. Hormel, Thomas S. Hwang, An-Lun Wu, Guangru B. Liang, **Yukun Guo**, Xiang Wei, Shuibin Ni, Yifan Jian, J. Peter Campbell, Steven T. Bailey, John C. Morrison and David Huang
Progress in Retinal and Eye Research, 2025. 107: 101378.

6. Glaucomatous Focal Perfusion Loss in the Macula Measured by Optical Coherence Tomographic Angiography
Aiyin Chen, Ping Wei, Jie Wang, Liang Liu, Acner Camino, **Yukun Guo**, Ou Tan, Yali Jia and David Huang
American Journal of Ophthalmology, 12, 2024. 268: 181-189.

7. Quantitative Evaluation of Type 1 and Type 2 Choroidal Neovascularization Components Under Treatment With Projection-Resolved OCT Angiography
Kotaro Tsuboi, Qi Sheng You, Jie Wang, **Yukun Guo**, Christina J. Flaxel, Thomas S. Hwang, David Huang, Yali Jia and Steven T. Bailey
Investigative Ophthalmology & Visual Science, 09, 2024. 65(11): 32-32.

8. Multi-Plexus Nonperfusion Area Segmentation in Widefield OCT Angiography Using a Deep Convolutional Neural Network
Yukun Guo, Tristan T. Hormel, Min Gao, Qisheng You, Jie Wang, Christina J. Flaxel, Steven T. Bailey, Thomas S. Hwang and Yali Jia
Translational Vision Science & Technology, 07, 2024. 13(7): 15-15.

9. Optimizing numerical k-sampling for swept-source optical coherence tomography angiography
Naixing Huang, Tristan T. Hormel, Guangru B. Liang, Xiang Wei, **Yukun Guo**, Siyu Chen and Yali Jia
Opt. Lett., Mar, 2024. 49(5): 1201--1204.

10. Visualizing features with wide-field volumetric OCT angiography
Tristan T. Hormel, Guangru B. Liang, Xiang Wei, **Yukun Guo**, Min Gao, Jie Wang, David Huang, Steven T. Bailey, Thomas S. Hwang and Yali Jia
Opt. Express, Mar, 2024. 32(6): 10329--10347.

11. Interpretable Diabetic Retinopathy Diagnosis based on Biomarker Activation Map
Pengxiao Zang, Tristan T. Hormel, Jie Wang, **Yukun Guo**, Steven T. Bailey, Christina J. Flaxel, David Huang, Thomas S. Hwang and Yali Jia
IEEE Transactions on Biomedical Engineering, 2023. 71(1): 14-25. PMID: 37405891

12. Single-shot OCT and OCT angiography for slab-specific detection of diabetic retinopathy
Guangru B Liang, Tristan T. Hormel, Xiang Wei, **Yukun Guo**, Jie Wang, Thomas S. Hwang and Yali Jia
Biomedical Optics Express, 2023. 14(11): 5682--5695. PMID: 38021127

13. Perfused and nonperfused microaneurysms identified and characterized by structural and angiographic optical coherence tomography
Min Gao, Tristan T. Hormel, **Yukun Guo**, Kotaro Tsuboi, Christina J. Flaxel, David Huang, Thomas S. Hwang and Yali Jia
Ophthalmology Retina, 2023. : S2468-6530(23)00438-4. PMID: 37673397

14. Early Sign of Retinal Neovascularization Evolution in Diabetic Retinopathy: A longitudinal OCT Angiography Study
Kotaro Tsuboi, Mehdi Mazloumi, **Yukun Guo**, Jie Wang, Christina J. Flaxel, Steven T. Bailey, David J. Wilson, David Huang, Yali Jia and Thomas S. Hwang
Ophthalmology Science, 2023. : 100382. PMID: 37868804

15. Utility of En face OCT for the Detection of Clinically Unsuspected Retinal Neovascularization in Patients with Diabetic Retinopathy
Kotaro Tsuboi, Mehdi Mazloumi, **Yukun Guo**, Jie Wang, Christina J. Flaxel, Steven T. Bailey, David Huang, Yali Jia and Thomas S. Hwang
Ophthalmology Retina, 2023. PMID: 36918122

16. Automated Macular Fluid Volume As a Treatment Indicator for Diabetic Macular Edema
Kotaro Tsuboi, Qisheng You, **Yukun Guo**, Jie Wang, Christina J. Flaxel, Steven T. Bailey, David Huang, Yali Jia and Thomas S. Hwang
Journal of VitreoRetinal Diseases, 2023. 7(3): 226-231. PMID: 37188216

17. Three-dimensional quantification of intraretinal cystoid spaces associated with full thickness macular hole
Kotaro Tsuboi, **Yukun Guo**, Jie Wang, Elizabeth White, Sam Mershon, Motohiro Kamei, David Huang, Yali Jia, Thomas S. Hwang and Steven T. Bailey
Retina (Philadelphia, Pa.), August, 2022. PMID: 36007173

18. A deep learning network for classifying arteries and veins in montaged wide-field OCT angiograms
Min Gao, **Yukun Guo**, Tristan T. Hormel, Kotaro Tsuboi, George Pacheco, David Poole, Steven T. Bailey, Christina J. Flaxel, David Huang, Thomas S. Hwang and Yali Jia

Ophthalmology Science, 4, 2022. : 100149. PMID: 36278031

19. Association Between Fluid Volume in Inner Nuclear Layer and Visual Acuity in Diabetic Macular Edema

Kotaro Tsuboi, Qisheng You, **Yukun Guo**, Jie Wang, Christina J. Flaxel, Steven T. Bailey, David Huang, Yali Jia and Thomas S. Hwang

American Journal of Ophthalmology, 12, 2021. 237: 164-172. PMID: 34942107

20. Geographic Atrophy Progression Is Associated With Choriocapillaris Flow Deficits Measured With Optical Coherence Tomographic Angiography

Qisheng You, Acner Camino, Jie Wang, **Yukun Guo**, Christina J. Flaxel, Thomas S. Hwang, David Huang, Yali Jia and Steven T. Bailey

Investigative ophthalmology & visual science, 12, 2021. 62(15): 28--28. PMID: 34964802

21. An Open-Source Deep Learning Network for Reconstruction of High-Resolution OCT Angiograms of Retinal Intermediate and Deep Capillary Plexuses

Min Gao, Tristan T. Hormel, Jie Wang, **Yukun Guo**, Steven T. Bailey, Thomas S. Hwang and Yali Jia

Translational Vision Science & Technology, 11, 2021. 10(13): 13-13. PMID: 34757393

22. Deep learning-based signal-independent assessment of macular avascular area on 6x6 mm optical coherence tomography angiogram in diabetic retinopathy: a comparison to instrument-embedded software

Honglian Xiong, Qisheng You, **Yukun Guo**, Jie Wang, Bingjie Wang, Liqin Gao, Christina J. Flaxel, Steven T. Bailey, Thomas S. Hwang and Yali Jia

British Journal of Ophthalmology, 9, 2021. PMID: 34518161

23. Normative intercapillary distance and vessel density data in the temporal retina assessed by wide-field spectral-domain optical coherence tomography angiography

Keke Liu, **Yukun Guo**, Qisheng You, Tristan T. Hormel, Thomas S. Hwang and Yali Jia

Experimental Biology and Medicine, 8, 2021. 246(20): 2230-2237. PMID: 34435914

24. An end-to-end network for segmenting the vasculature of three retinal capillary plexuses from OCT angiographic volumes

Yukun Guo, Tristan T. Hormel, Shaohua Pi, Xiang Wei, Min Gao, John C. Morrison and Yali Jia

Biomed. Opt. Express, 8, 2021. 12(8): 4889--4900. PMID: 34513231

25. Comparison of Central Macular Fluid Volume With Central Subfield Thickness in Patients With Diabetic Macular Edema Using Optical Coherence Tomography Angiography

Qisheng You, Kotaro Tsuboi, **Yukun Guo**, Jie Wang, Christina J. Flaxel, Steven T. Bailey, David Huang, Yali Jia and Thomas S. Hwang

JAMA Ophthalmology, 7, 2021. 139(7): 734-741. PMID: 33983385

26. Quantification of Nonperfusion Area in Montaged Widefield OCT Angiography Using Deep Learning in Diabetic Retinopathy

Yukun Guo, Tristan T. Hormel, Liqin Gao, Qisheng You, Jie BingWang, Christina J. Flaxel, Steven T. Bailey, Dongseok Choi, David Huang, Thomas S. Hwang and Yali Jia

Ophthalmology Science, 6, 2021. 1(2): 100027. PMID: 36249293

27. Plexus-specific retinal capillary avascular area in exudative age-related macular degeneration with projection-resolved OCT angiography

Liqin Gao, Jie Wang, Qisheng You, **Yukun Guo**, Christina J. Flaxel, Thomas S. Hwang, David Huang, Yali Jia and Steven T. Bailey

British Journal of Ophthalmology, 12, 2020. PMID: 33355172

28. Automated Segmentation of Retinal Fluid Volumes From Structural and Angiographic Optical Coherence Tomography Using Deep Learning

Yukun Guo, Tristan T. Hormel, Honglian Xiong, Jie Wang, Thomas S. Hwang and Yali Jia

Translational Vision Science & Technology, 10, 2020. 9(2): 54. PMID: 33110708

29. Optical Coherence Tomography Angiography Avascular Area Association With 1-Year Treatment Requirement and Disease Progression in Diabetic Retinopathy

Qisheng You, Jie Wang, **Yukun Guo**, Shaohua Pi, Christina J. Flaxel, Steven T. Bailey, David Huang, Yali Jia and Thomas S. Hwang

American Journal of Ophthalmology, 9, 2020. 217: 268--277. PMID: 32360332

30. Reconstruction of high-resolution 6×6-mm OCT angiograms using deep learning

Min Gao, **Yukun Guo**, Tristan T. Hormel, Jiande Sun, Thomas S. Hwang and Yali Jia

Biomedical Optics Express, 7, 2020. 11(7): 3585--3600. PMID: 33014553

31. High-resolution wide-field OCT angiography with a self-navigation method to correct microsaccades and blinks

Xiang Wei, Tristan T. Hormel, **Yukun Guo**, Thomas S. Hwang and Yali Jia

Biomedical Optics Express, 6, 2020. 11(6): 3234--3245. PMID: 32637251

32. Depth-resolved optimization of a real-time sensorless adaptive optics optical coherence tomography

Acner Camino, Ringo Ng, Joey Huang, **Yukun Guo**, Shuibin Ni, Yali Jia, David Huang and Yifan Jian

Opt. Lett., May, 2020. 45(9): 2612--2615. PMID: 32356829

33. Detection of Clinically Unsuspected Retinal Neovascularization with Wide-Field Optical Coherence Tomography Angiography

Qisheng You, **Yukun Guo**, Jie Wang, Xiang Wei, Acner Camino, Pengxiao Zang, Christina J. Flaxel, Steven T. Bailey, David Huang, Yali Jia and Thomas S. Hwang

Retina (Philadelphia, Pa.), 5, 2020. 40(5): 891--897. PMID: 30845022

34. Automated diagnosis and segmentation of choroidal neovascularization in OCT angiography using deep learning

Jie Wang, Tristan T. Hormel, Liqin Gao, Pengxiao Zang, **Yukun Guo**, Xiaogang Wang, Steven T. Bailey and Yali Jia

Biomedical Optics Express, 2, 2020. 11(2): 927. PMID: 32133230

35. Robust non-perfusion area detection in three retinal plexuses using convolutional neural network in OCT angiography

Jie Wang, Tristan T. Hormel, Qisheng You, **Yukun Guo**, Xiaogang Wang, Liu Chen, Thomas S. Hwang and Yali Jia

Biomedical Optics Express, 1, 2020. 11(1): 330--345. PMID: 32010520

36. Detection of Reduced Retinal Vessel Density in Eyes with Geographic Atrophy Secondary to Age-Related Macular Degeneration Using Projection-Resolved Optical Coherence Tomography Angiography

Qisheng You, Jie Wang, **Yukun Guo**, Christina J. Flaxel, Thomas S. Hwang, David Huang, Yali Jia and Steven T. Bailey

American Journal of Ophthalmology, 1, 2020. 209: 206--212. PMID: 31526797

37. 75-degree non-mydriatic single-volume optical coherence tomographic angiography

Xiang Wei, Tristan T. Hormel, **Yukun Guo** and Yali Jia

Biomedical Optics Express, 12, 2019. 10(12): 6286. PMID: 31853400

38. Development and validation of a deep learning algorithm for distinguishing the nonperfusion area from signal reduction artifacts on OCT angiography

Yukun Guo, Tristan T. Hormel, Honglian Xiong, Bingjie Wang, Acner Camino, Jie Wang, David Huang, Thomas S. Hwang and Yali Jia

Biomed. Opt. Express, 7, 2019. 10(7): 3257--3268. PMID: 31360599

39. Three-dimensional structural and angiographic evaluation of foveal ischemia in diabetic retinopathy: method and validation

Bingjie Wang, Acner Camino, Shaohua Pi, **Yukun Guo**, Jie Wang, David Huang, Thomas S. Hwang and Yali Jia

Biomedical Optics Express, 7, 2019. PMID: 31360604

40. High dynamic range optical coherence tomography angiography (HDR-OCTA)

Xiang Wei, Tristan T. Hormel, Shaohua Pi, **Yukun Guo**, Yifan Jian and Yali Jia

Biomedical Optics Express, 7, 2019. PMID: 31360605

41. Monitoring retinal responses to acute intraocular pressure elevation in rats with visible light optical coherence tomography

Shaohua Pi, Tristan T. Hormel, Xiang Wei, William Cepurna, Acner Camino, **Yukun Guo**, David Huang, John C. Morrison and Yali Jia

Neurophotonics, 7, 2019. 6(4): 1--10. PMID: 31312671

42. Detecting and measuring areas of choriocapillaris low perfusion in intermediate, non-neovascular age-related macular degeneration

Acner Camino, **Yukun Guo**, Qisheng You, Jie Wang, David Huang, Steven T. Bailey and Yali Jia

Neurophotonics, 2019. 6(4): 1 -- 9. PMID: 31528658

43. MEDnet, a neural network for automated detection of avascular area in OCT angiography

Yukun Guo, Acner Camino, Jie Wang, David Huang, Thomas S. Hwang and Yali Jia

Biomedical Optics Express, 11, 2018. 9(11): 5147. PMID: 30460119

44. Automated segmentation of retinal layer boundaries and capillary plexuses in wide-field optical coherence tomographic angiography

Yukun Guo, Acner Camino, Miao Zhang, Jie Wang, David Huang, Thomas S. Hwang and Yali Jia

Biomedical Optics Express, 9, 2018. 9(9): 4429. PMID: 30615747